Forecasting Ottawa's Electricity Load for a Smarter Grid

Weather- and calendar-aware ML forecasts of hourly demand and baseload, with a net-load view of solar, wind and EV fleets

Kshitij · Victor

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Problem: what drives Ottawa's electricity load

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Weather
Cold winters drive heating peaks; hot summers drive cooling peaks



Calendar
Workdays, weekends and statutory holidays change the daily shape



Renewables
Solar and wind output vary hour to hour with the forecast



Electrification
EV fleets add large, schedulable charging loads

Research question

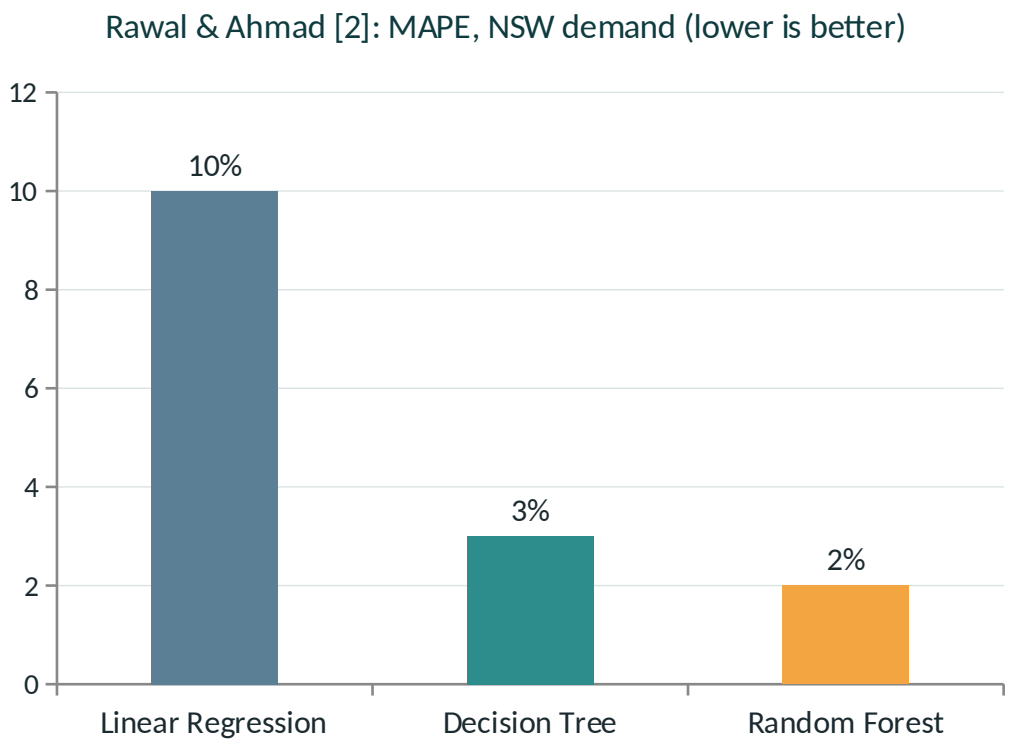
Can tree-ensemble models (Random Forest, XGBoost) forecast Ottawa's hourly electricity demand and next-day baseload more accurately than a Linear Regression baseline, using lagged demand, weather and calendar features?

Why it matters: a good forecast lets the grid schedule supply, storage and flexible loads (like EV charging) ahead of demand instead of reacting to it.

Baseload = the minimum level of demand the grid must always serve (we use the daily minimum hourly load).

State of the art: tree ensembles beat linear models

Victor



Same study: R^2 0.31 (LR) → 0.93 (DT) → 0.97 (RF); MAE 802 → 220 → 159 MW.

[1] Gökçe & Duman, IEEE IISEC 2022
Hourly Turkish demand with temperature, holidays and a 24 h lag. RF and XGBoost beat Linear Regression; RF had the lowest error.

[2] Rawal & Ahmad, IEEE ICPC2T 2022
Half-hourly NSW (Australia) demand, 2020. Non-parametric models beat parametric ones; Random Forest was best.

[3] Kong et al., IEEE Trans. Smart Grid 2019
LSTM recurrent networks for short-term residential load. Deep sequence models win at noisy, household-level scale.

Gap we address: these studies forecast a whole country or state. None targets a single city zone like Ottawa, or asks what renewables and EV fleets do to the net load that remains to be served.

Proposed solution: forecast, then balance

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1 · Real data

IESO hourly demand (Ottawa zone), ECCC + Open-Meteo weather, Ontario holidays

2 · Features

Lags t-1, t-24, t-168; temperature, HDD/CDD; hour, weekday, month, holiday flag

3 · ML models

Linear Regression (baseline), Random Forest, XGBoost

4 · Forecasts

Next-hour demand and next-day baseload (daily minimum)

5 · Scenario layer: live supply-demand view

Net load = forecast demand − solar farm − wind farm + EV fleet charging

Solar and wind come from weather forecasts through physical models; EV charging from workday and holiday schedules. The result shows, hour by hour, how much dispatchable supply is needed and when flexible EV charging can be shifted.

Data: real where possible, assumed where not

Victor

Component	Source	Resolution	Type
Ottawa-zone electricity demand (target)	IESO Hourly Zonal Demand Report, Ottawa zone [4]	Hourly, 2020–2025	**REAL**
Temperature, humidity	ECCC historical climate data, Ottawa station [5]	Hourly	**REAL**
Solar irradiance, 100 m wind speed	Open-Meteo historical weather API [6]	Hourly	**REAL**
Holidays, workdays	Ontario statutory holiday calendar	Daily	**REAL**
Solar farm output	pvlib PV model [7] driven by real irradiance	Hourly	**ASSUMED**
Wind farm output	Generic turbine power curve driven by real wind speed	Hourly	**ASSUMED**
Residential vs. industrial split, EV fleet load	Calendar-based profiles (next slide)	Hourly	**ASSUMED**

Split: train 2020–2023 · validate 2024 · test 2025 (chronological, no shuffling, so no future data leaks into training).

Scenario assumptions: explicit and adjustable

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Solar farm

10 MW AC fixed-tilt array near Ottawa. Output from real irradiance and temperature through pvlib; zero at night, low under winter snow.

Wind farm

20 MW (e.g. 6 × 3.3 MW). Power curve: cut-in 3 m/s, rated 12 m/s, cut-out 25 m/s, from 100 m forecast wind speed.

Residential

Follows temperature (heating and cooling degree-hours). Morning and evening peaks; higher on weekends and holidays.

Industrial / commercial

Flat daytime block on workdays, reduced on weekends and statutory holidays, minimal overnight.

EV fleet

300 fleet vehicles, ~60 kWh each per workday, charged overnight (22:00–06:00); ~30% of that on holidays.

Calibration

Shares are scaled so their sum matches the real IESO Ottawa-zone total. Every parameter is a variable we can stress-test.

Methodology: models, baselines and evaluation

Victor

Persistence / seasonal naive
Floor every model must beat: demand(t+1) = demand(t), or same hour last week

Linear Regression
Interpretable baseline; shows how much is linear in temperature and calendar

Random Forest
Bagged trees; captures nonlinear temperature response (U-shaped heating/cooling)

XGBoost
Gradient-boosted trees; strongest model in the literature for tabular load data [1]

Stretch goal: LSTM sequence model [3] if time allows.

Evaluation on the unseen 2025 test year

MAE MW	**RMSE** MW
MAPE %	**R^2** fit

Critical analysis: errors by season, peak hours and holidays; feature importance; residual plots. Then net-load error after adding the solar, wind and EV scenarios.

Feasibility and timeline

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Why this scope is realistic

- Data is free, public and hourly: ~52,000 rows for 2020–2025, which trains in seconds to minutes on a laptop.
- Standard, well-documented libraries: pandas, scikit-learn, XGBoost, pvlib.
- The core ML deliverable (steps 1–4) does not depend on the scenario layer, so the scenarios can't put it at risk.
- Clear fallback: if zonal data is incomplete, we use IESO Ontario-wide hourly demand with the same pipeline.
- Out of scope: real-time control, grid power-flow simulation, private utility data.

Oct 13
Proposal (this talk)

Oct 14–31
Collect and clean data; feature pipeline; EDA

Nov 1–15
Train baselines, RF and XGBoost; tune on 2024

Nov 16–28
Scenario layer: solar, wind, EV, net load

Nov 29–Dec 8
Test-year analysis; final presentation

Dec 14
IEEE-format final report

Team contributions and use of AI

Kshitij
Problem definition and research question; smart-grid scenario layer (solar, wind, EV, net load); feasibility and timeline; slide design.

Victor
State-of-the-art review; data sources and preprocessing pipeline; model training and evaluation design.

Both members will share coding and writing of the final report.

AI disclosure

ChatGPT (OpenAI): summary tables and illustrative diagrams in our draft.

Claude (Anthropic), via Claude Code: restructuring the deck for the 5-minute limit, slide wording and layout, checking citations, and drafting the scenario assumptions for us to review.

Our own work: choice of problem, papers, data sources and methods. We checked every number against the cited papers and are responsible for all content.

References

[1] Gökçe and Duman, “Performance Comparison of Simple Regression, Random Forest and XGBoost Algorithms for Forecasting Electricity Demand,” in Proc. 3rd Int. Informatics and Software Engineering Conf. (IISEC), Ankara, 2022. doi: 10.1109/IISEC56263.2022.9998213.

[2] Rawal and Ahmad, “A Comparative Analysis of Supervised Machine Learning Algorithms for Electricity Demand Forecasting,” in Proc. 2nd Int. Conf. Power, Control and Computing Technologies (ICPC2T), 2022, pp. 1–6. doi: 10.1109/ICPC2T53885.2022.9776960.

[3] W. Kong, Z. Y. Dong, Y. Jia, D. J. Hill, Y. Xu and Y. Zhang, “Short-Term Residential Load Forecasting Based on LSTM Recurrent Neural Network,” IEEE Trans. Smart Grid, vol. 10, no. 1, pp. 841–851, 2019. doi: 10.1109/TSG.2017.2753802.

[4] Independent Electricity System Operator (IESO), “Hourly Zonal Demand Report.” [Online]. Available: https://reports-public.ieso.ca/public/DemandZonal/

[5] Environment and Climate Change Canada, “Historical Climate Data.” [Online]. Available: https://climate.weather.gc.ca/

[6] Open-Meteo, “Historical Weather API.” [Online]. Available: https://open-meteo.com/en/docs/historical-weather-api

[7] W. F. Holmgren, C. W. Hansen and M. A. Mikofski, “pvlib python: a python package for modeling solar energy systems,” J. Open Source Software, vol. 3, no. 29, p. 884, 2018. doi: 10.21105/joss.00884.



Questions?

Forecast Ottawa's load from real data · balance it against solar, wind and EV fleets